

Measurement System Assessment (Simple Gage R&R)

What We Measured

Measurement: checkout waiting time, in minutes, from joining the line until payment complete. Two cashiers both recorded times for the same 10 customers to see if their measurements were similar.

Small Trial Results (Illustrative)

Customer	Cashier A Time (min)	Cashier B Time (min)	Difference (A - B)
1	8.0	8.2	-0.2
2	7.5	7.4	0.1
3	9.0	9.1	-0.1
4	6.8	6.9	-0.1
5	10.0	9.9	0.1
6	8.5	8.4	0.1
7	7.2	7.3	-0.1
8	11.0	11.1	-0.1
9	5.5	5.6	-0.1
10	9.5	9.4	0.1

Simple Conclusion

- The differences between the two cashiers are very small (around 0.1–0.2 minutes).
- For this project, that level of variation is acceptable.
- A full statistical Gage R&R study could be done, but for a simple timing task this quick check is enough.